

JobLaunch Project – Test Plan

Introduction

This Test Plan outlines the strategy and scope for testing the JobLaunch web-based job application system.

The goal is to ensure that all functional and non-functional requirements defined in the SRS and Design documents are implemented correctly, reliably, and securely.

Objectives

- ✓ Verify that all core features (Authentication, Job Posting, Search, Application Workflow) function as intended.
- ✓ Validate non-functional requirements (performance, security, usability).
- ✓ Identify and document defects, ensuring they are resolved before final deployment.
- ✓ Provide confidence in the stability of the prototype for the final demo.

Scope

In-Scope:

- ✓ Functional testing of all features defined in the SRS.
- ✓ Non-functional testing: performance, security, usability.
- ✓ End-to-end workflow testing (Seeker applies → Employer updates status → Seeker views status).

Out-of-Scope:

- ✓ Advanced scalability features (Redis caching, Elasticsearch).
- ✓ Third-party integration beyond MVP (resume parsing, real-time notifications).

Test Items

- ✓ Front-end UI (Sign-up, Login, Job Listings, Application forms).
- ✓ Back-end API s (Auth, Job CRUD, Application submission/status).
- ✓ Database (Users, Jobs, Applications tables).
- ✓ Deployment environment (Vercel + Supabase).

Test Strategy

JobLaunch Project – Test Plan

- ✓ Unit Testing: Developers test individual functions (e.g., login, createJob).
- ✓ Integration Testing: QA verifies API + DB interactions (e.g., job creation stored in DB).
- ✓ System Testing: End-to-end workflows tested by QA.
- ✓ User Acceptance Testing (UAT): Simulate real user scenarios (Seeker applying, Employer updating status).

Test Environment

- ✓ Frontend deployed on Vercel (staging).
- ✓ Back-end server-less functions on Vercel.
- ✓ Supabase PostgreSQL database.
- ✓ Tools: Postman (API testing), Jest/Mocha (unit tests), Browser DevTools (UI testing).

Entry and Exit Criteria

Entry: Code deployed to staging, core features implemented.

Exit: All critical test cases passed, major bugs resolved, prototype stable.

Deliverables

- ✓ Test Plan (this document).
- ✓ Test Case Document (detailed cases for each feature).
- ✓ Bug Report Log (issues and resolutions).
- ✓ Evidence of Testing (screenshots, logs).

Roles and Responsibilities

- ✓ QA Lead (Netsanet Haile): Prepare Test Plan, write Test Cases, execute tests, log bugs.
- ✓ Developers (Julas, Tsgabu): Fix defects, provide unit test coverage.
- ✓ Project Manager (Mekbb): Coordinate timelines, ensure deliverables are submitted.
- ✓ Documentation Lead (Faniel): Maintain testing artifacts in GitHub repo.

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Problem Area	Specific Problem	Testing Tool(s)	How to Fix the Problem
Unit Testing	Empty job title accepted	Jest / JUnit / Mocha / PyTest	Add input validation to reject empty fields and return error messages
Unit Testing	Weak or short password allowed	Jest / JUnit / Mocha / PyTest	Enforce password length & complexity rules in backend validation
Unit Testing	Job CRUD not working correctly	Jest / JUnit / Mocha / PyTest	Debug CRUD functions, ensure correct database queries
Integration Testing	Front-end not communicating with API	Postman / Newman	Check API endpoints, HTTP methods, and request payloads
Integration Testing	Data not saved to database	Postman / Cypress	Verify database connection, ORM queries, and error handling
Integration Testing	Job application not stored	Cypress / Selenium	Fix API logic and ensure transaction commits
System Testing	Full user workflow fails	Selenium / Cypress	Identify broken steps and fix routing, state management
System Testing	Website not responsive on mobile	Lighthouse / Cypress	Apply responsive CSS
System Testing	Browser compatibility issues	Selenium	Fix CSS/JS issues specific to browsers (Chrome, Firefox, Edge)
UAT (User Acceptance Testing)	User confused during registration	Manual Testing	Improve UI/UX, add hints, simplify forms
UAT	Employer takes too long to post job	Manual Testing / Feedback	Reduce steps, optimize form flow
Test Case Documentation	Missing test cases for critical features	Test Case Documents	Write detailed test cases for login, job post, search, apply
Security	Unauthorized access to admin pages	Manual / Integration Tests	Implement role-based access control (RBAC)
Performance	Slow page load	Lighthouse	Optimize images, minimize JS/CSS, use caching